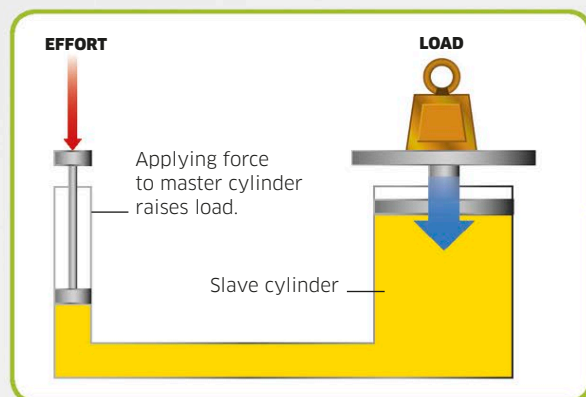


# Simple machines

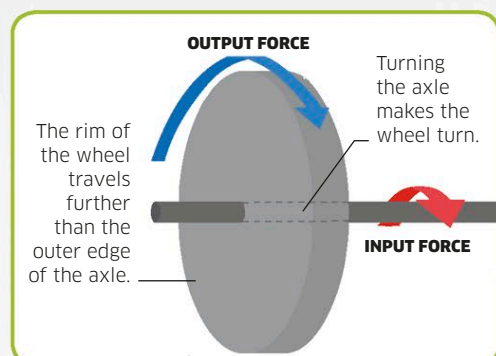
A machine is anything that changes the size or direction of a force, making work easier. Simple machines include ramps, wedges, screws, levers, wheels, and pulleys.

Complex machines such as cranes and diggers combine a number of simple machines, but whatever the scale, the physical principles remain the same. Many of the most effective machines are the simplest—a sloping path (ramp); a knife (wedge); a jar lid (screw); scissors, nutcrackers, and tweezers (levers); a faucet (wheel and axle); or hoist (pulley), for example. Hydraulics and pneumatics use the pressure in fluids (liquids and gases) to transmit force.



## Hydraulics

A hydraulic system makes use of pressure in a liquid by applying force (effort) to a “master” cylinder, which increases fluid pressure in a “slave” cylinder. The hydraulic ram lifts the crane’s boom by using pressure from fluid in the cylinder to push a piston.



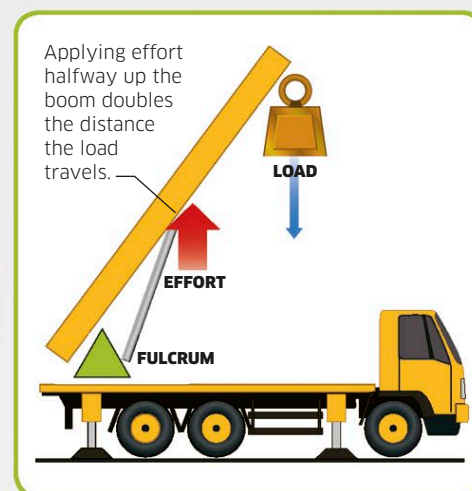
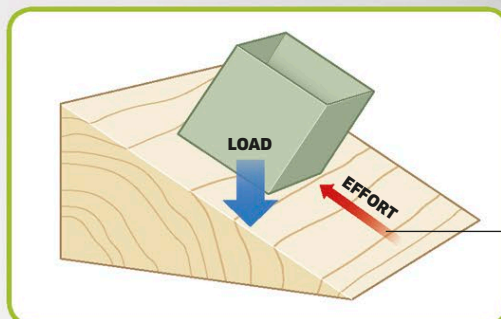
## Wheel and axle

A wheel with an axle can be used in two ways: either by applying a force to the axle to turn the wheel, which multiplies the distance traveled; or by applying a force to the wheel to turn the axle, like a spanner.

## Ramp and wedge

Also known as an inclined plane, a ramp reduces the force needed to move an object from a lower to a higher place.

A wedge acts like a moving inclined plane, applying a greater force to raise an object.



## Lever

The crane’s boom is a long, third-class lever. When a hydraulic ram applies a force greater than the load between the load and the fulcrum, the crane lifts the load.

Less effort is needed to push a load up a ramp, but the load has to move further along the slope than it moves vertically.